



FLEET & SHORE PERSONNEL TRAINING

Vessel General Permit

Managing incidental discharges from normal vessel operations in United States waters.



US EPA 2013 VGP

CWA 502

40 CFR 122.2

33 CFR 151 / 155

What we will cover

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SECTION 01

Understanding the VGP

The Vessel General Permit governs the ordinary, incidental discharges that every ship produces — not accidents and not cargo residues.

01

What the Vessel General Permit is

The VGP covers commercial ships and large pleasure craft operated in United States waters.



A discharge permit

It authorises the conventional, incidental discharges of normal vessel operation — machinery cooling water, rainwater collecting on deck, deck washdown water and similar.



Permitted waters

United States territorial waters as defined by CWA 502 (as defined in 40 CFR 122.2): inland waters and waters reaching 3 miles offshore, from CWA 502(8).



Twenty-six categories

The US Environmental Protection Agency has identified 26 discharge operations carried out on ships under the VGP.



What sits outside it

Some discharges are not covered by the VGP at all; others are covered by different US regulations — for example sewage, oil and petroleum wastes.

KEY NUMBERS

The figures that define the permit

Four numbers govern almost every VGP conversation on board.

26

Discharge categories identified
by the EPA under the VGP

3

Miles offshore — the seaward
edge of permitted waters

3

Years that VGP records must be
retained on board

30

Days between submitting the
NOI and coverage taking effect



All US waters — Including the Pacific and the Atlantic coasts, Hawaii and Puerto Rico, and all inland waters.



No modification needed — Technically the VGP requires no ship modification and no additional equipment — it requires practice and maintenance.

PURPOSE

The aim of the VGP

The purpose of the permit is the cleanest possible discharges to the sea, through effective management practices (BMP) and sound maintenance.



Minimise discharges to the sea

Where it is possible to reduce or eliminate a discharge, that is the required outcome — not merely staying inside a numerical limit.



Follow maintenance procedures

Ensure that planned maintenance and good-housekeeping procedures are actually followed and are seen to be followed.



No oil and no waste on deck

The deck must be free of any residue that could reach the sea with rain, spray or washing water.



Use effective management practices

Bring each discharge to an acceptable level using available technology and the best marine practice within the VGP limits.



Nothing to install

The VGP is not an equipment standard. It asks the crew to operate and maintain what is already fitted so that discharges to the sea are as clean and as small as they can be made.

The four standing VGP requirements

Whatever the trade, every covered vessel owes the EPA these four things.



Self-audits

Routine visual inspections and a comprehensive annual audit of the vessel.



Record keeping

A complete VGP record, retained on board for three years.



Training

Documented familiarisation for every person whose work touches a covered discharge.



Reporting

Non-conformities reported to the relevant EPA regional office.

A POINT THAT IS OFTEN MISSED

A leak from poor maintenance is an offence, not an accident

A leak arising from the regular maintenance of the ship or its equipment is not considered acceptable even where the substance itself is not covered by this permit — it is treated as an offence.

Maintain it

Contain it

Record it

Covered by the VGP, or covered by something else

Knowing which regime applies decides where the waste goes and which record it lands in.



Covered by the VGP

26 incidental discharges

- Deck washdown water and rainwater puddles
- Bilgewater from a correctly functioning oily water separator
- Ballast water and chain locker effluent
- Machinery and seawater cooling discharges
- Graywater from accommodation spaces



Covered by other US rules

Different permits apply

- Sewage from vessels
- Oil and petroleum wastes
- Garbage and used oils
- Medical waste and chemical liquid wastes
- Discharges permitted under other regulations



Prohibited outright

No permit exists

- Discharge of toxic and hazardous substances to the environment
- Tetrachloroethylene (perchloroethylene) degreaser
- Photo processing and dry cleaning wastes
- Any discharge causing a visible sheen

Which ships, which waters, and when



All commercial vessels and cruise vessels

The permit binds the owner or operator of every commercial ship or cruise vessel operating in US territorial waters, including Hawaii and Puerto Rico.



Coastline to 3 miles, plus inland waters

In general US territorial waters cover the distance from the coastline out to 3 miles, together with all inland waters. The permit is valid only in those waters.



Frequent callers comply continuously

Ships that call at US ports frequently should meet the requirements of this procedure at all times.



Occasional callers comply before arrival

Ships that rarely call at US ports will meet the requirements of this procedure before arrival at the port.

What every crew member needs to know

All personnel of a ship operating in US territorial waters must be trained to become familiar with the VGP, and that training must be recorded.



The requirements for your own workspace

Everyone has to know all of the VGP requirements that apply to the spaces and equipment they work with.



How to report a violation

Everyone has to know the responsibilities that arise from reporting violations of the permit.



How to report a sheen on the water

Everyone has to know the duty to report a sheen or glow on the water surface caused by a leak.



That the training is recorded

Familiarisation is a documented requirement — an untrained crew is itself a non-conformity.



It has to be written down

VGP training is verified from records. If the familiarisation is not entered in the training record, an inspector will treat it as if it never happened.

What minimise actually requires

Within the VGP limits, minimise means bringing a discharge to an acceptable level, or eliminating it, by taking the necessary measures and using effective management practices (BMP) within available technology and best marine practice.



Acceptable

- Reduce the volume discharged wherever the ship has the capability
- Hold the effluent on board and land it ashore where a tank is available
- Use available technology and best marine practice to clean the discharge
- Choose the position and moment that cause the least environmental impact
- Record what was discharged, when, where and how much



Never acceptable

- Use additives during normal vessel discharge operations
- Dilute the discharged fluid in order to meet a numerical limit
- Treat a numerical limit as a target rather than a ceiling
- Discharge simply because no limit is written for that operation
- Leave a discharge unrecorded because it appeared harmless

Discharges specific to ordinary ship operations

These are the everyday operations the permit was written for.



Deck washdown water

Water accumulating on the deck after the deck has been washed, or after rain and spray.



Bilgewater via the OWS

Bilgewater produced by a properly functioning oily water separator device.



Ballast water

Ballast water taken, exchanged and discharged in the course of normal trading.



SECTION 02

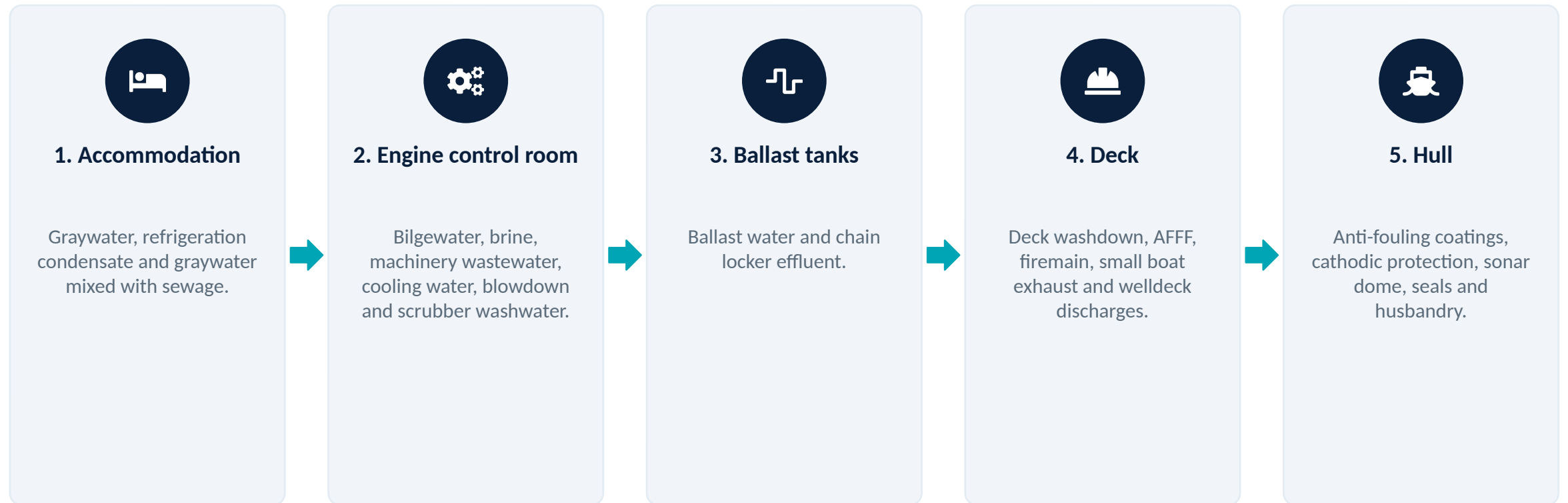
Discharges Covered by the Permit

Twenty-six categories, each with its own paragraph number in VGP Part 2.2. This section follows them area by area.

02

The categories, grouped by where they arise

Every one of the 26 VGP categories belongs to a space someone is responsible for. Learn the ones that belong to yours.



LIMITS

Distance and concentration limits to memorise

Where the VGP or Company procedure sets a figure, the figure is a ceiling — not a target.

Ref	Discharge	Limit that applies
2.2.15	Graywater	Only while under way and a minimum of 1 mile offshore; never if the ship can hold it in its own tank
2.2.8	Chain locker effluent	Anchor and chain washed after each use; chain locker pumped only a minimum of 50 miles offshore
2.2.19	Seawater cooling overboard	Minimise; cooling water strainers and filters cleaned a minimum of 50 miles from land
2.2.20	Seawater piping biofouling prevention	Fouling organisms cleaned from the piping a minimum of 50 miles offshore; FIFRA-approved chemicals only
2.2.2	Bilgewater	Never in US waters except in an emergency; no residue or sheen on the water surface
2.2.16	Motor gasoline and compensating discharge	Never overboard in VGP waters; routed to the bilge tank and discharged below 15 ppm through the OWS

All distances are from the nearest land unless the paragraph states otherwise.

Definitions taken from 40 CFR 1700.4

Six of the categories carry a formal definition. Using the correct wording keeps the record defensible.

Ref	Term	Definition
2.2.10	Distillation and reverse osmosis brine	Concentrated seawater produced by the system used to turn seawater into fresh water.
2.2.11	Elevator pit effluent	The water collected and discharged from the pit at the bottom of the vessel elevator shaft.
2.2.5	Aqueous film-forming foam (AFFF)	A fire-fighting substance mixed with seawater, which may also be used in training, testing and maintenance.
2.2.22	Sonar dome discharge	Seawater or fresh water held in the sonar dome, and the washing of the anti-foulant on the dome.
2.2.24	Welldeck discharges	Discharge of residues from the welldeck — the opening at the waterline where small and amphibious craft dock and are supplied.
2.2.23	Underwater ship husbandry	Discharge of the wastes generated while cleaning the hull and underwater body of the ship in the water.

Graywater

Graywater covers the water from galley sinks, baths, showers, washbasins, laundries and drinking fountains.



Minimise the discharge

All ships will minimise graywater discharge as far as it is possible to do so.



Hold it in the tank

If the ship has the capability, graywater will be held in the ship graywater tank.



Under way and 1 mile off

Ships able to store graywater in their own tank may discharge it while the ship is under way and a minimum of 1 mile offshore.



Phosphate-free products only

Soaps and detergents free of phosphate and toxic substances must be used. This has to be considered at the requisition stage.



Keep cooking fats out

Fats used in cooking must never be drained into the graywater system. Food and oil residues are cleaned from the dishes as far as possible before washing.

Refrigeration condensate and graywater mixed with sewage



2.2.18 Refrigeration and air condensate

Prohibited

- Discharge of condensate from refrigeration plant, freezers and air conditioning condensers
- Discharge of these substances is prohibited in US territorial waters
- Route the condensate to a holding tank and land it ashore



2.2.25 Graywater containing sewage

Emergencies only

- Graywater that contains sewage must not be discharged except in an emergency
- Sewage itself is not covered by the VGP — other US regulations apply
- Any emergency discharge must be recorded

Bilgewater



Never in US waters

Bilgewater must never be discharged in US territorial waters except in an emergency.



Cargo space bilges

Except in exceptional circumstances, bilge water in cargo spaces must never be discharged to the sea.



No residue and no sheen

If a discharge operation is unavoidable, the liquid pumped during the operation must leave no residue and no sheen on the water surface.



Record every operation

Every discharge operation is recorded, with position, time and quantity.



The OWS is the gate

Where discharge is permitted at all, it is only through a correctly functioning oily water separator, below 15 ppm, in waters where MARPOL permits it — otherwise the effluent is landed to a shore station.

Brine and elevator pit effluent



2.2.10 Distillation and reverse osmosis brine

Keep it clean at source

- Concentrated seawater produced as a waste by the freshwater system
- Must never come into contact with machinery or industrial equipment
- Must never come into contact with toxic or harmful substances or waste
- Definition source: 40 CFR 1700.4



2.2.11 Elevator pit effluent

To the bilge tank

- Overboard discharge must never be made except in an emergency
- The collected water is transferred to the engine room bilge tank
- It is then delivered ashore via the oily water separator
- All discharge operations must be recorded

Freshwater layup, gas turbine wash water, compensating discharge



2.2.13 Freshwater layup

Substances used as disinfectants in the water must be used at the minimum level needed to prevent aquatic life forming in the system.



2.2.14 Gas turbine wash water

Water used for washing gas turbines must never be discharged overboard into the sea.



2.2.16 Motor gasoline and compensating discharge

Seawater used to prevent vapour forming in fuel tanks and to fill the empty part of the tank. It can never be discharged in VGP waters — it goes to the bilge tank and out below 15 ppm through the OWS.

Machinery wastewater and seawater cooling

Non-oily machinery wastewater covers water from distillation plant, cooling devices, water pipes and valves, air compressors, and seawater and freshwater portable pumps.



Oil-free wastewater may be discharged

2.2.17 — machinery wastewater that contains no oil may be discharged to sea if the necessary precautions are taken to prevent it mixing with oils and additives.



But treat it as potentially oily

Since all machinery wastewater can potentially contain oil, it should be discharged through the OWS or landed to a shore station in areas permitted under MARPOL rules.



Cooling water is minimised

2.2.19 — discharge of seawater cooling water not in contact with machinery or oil, including water used to cool hydraulic systems, is minimised as far as possible.



Strainers cleaned 50 miles out

Cooling water strainers and filters are cleaned frequently, and that work is done a minimum of 50 miles from land.



Take shore power where fitted

Where suitable systems are available, the ship should take its energy supply from the port rather than run machinery alongside.

Seawater piping biofouling prevention

Seawater discharge containing additives used to prevent the formation and growth of biofouling organisms in seawater cooling systems.

- Biofouling chemicals used in seawater piping must be FIFRA approved.
- If a product is prohibited in the USA, discharge of that product is prohibited.
- To keep fouling under control, ships will use as little chemical as the job allows.
- Ships should regularly clean fouling organisms from the seawater piping.
- That cleaning operation must be performed a minimum of 50 miles offshore.
- Keep the product data sheet and the FIFRA approval with the VGP records.



FIFRA approval, on board

FIFRA is the US Federal Insecticide, Fungicide and Rodenticide Act. An inspector will ask to see that the biocide in use is approved for use in the United States before asking anything about the discharge itself.

Boiler blowdown and exhaust gas scrubber washwater



2.2.6 Boiler / economizer blowdown

Prohibited except in emergency

- Except in an emergency, any discharge from the boiler is prohibited
- Plan blowdown for waters outside the permit area
- Record any emergency discharge with reason, time and position



2.2.26 Exhaust gas scrubber washwater

Low sulphur fuel first

- Low sulphur fuels should be used in US territorial waters
- Scrubber washwater residues must not contain oil or oily mixtures in any appreciable quantity
- Sludge created by the scrubber must not be discharged in US territorial waters

Ballast water and chain locker effluent



2.2.3 Ballast water

See section 05 for the detail

- The ballast water record book must be kept correctly and in good order
- A ballast water exchange must be carried out to meet the additional VGP requirements for the voyage
- Note: USCG requirements, 33 CFR 155



2.2.8 Chain locker effluent

Sediment stays out of US waters

- Anchor and chain are washed down after each use
- The chain locker is pumped out only a minimum of 50 miles offshore
- Chains and lockers are cleaned and maintained while the ship is in the shipyard
- A certificate or written declaration covering that shipyard cleaning must be held in the shipyard report

Deck washdown, runoff and above-waterline cleaning

The deck must be free of every residue that could go overboard with rain, spray or washing water.

✓ Required

- Keep the deck clean and in good condition at all times
- Fit scupper plugs before cargo operations begin
- Clean every oil spill on deck with suitable cleaning materials
- Observe the water around the ship continuously for leaks
- Train personnel in the environmental protection procedures

⊘ Prohibited in US waters

- Wash decks in US territorial waters
- Allow residues to remain where rain or spray can carry them overboard
- Hose a spill overboard instead of collecting it
- Leave loose material on deck before a passage in heavy weather
- Assume rainwater runoff is clean without checking it

Aqueous film-forming foam (AFFF)

AFFF is a fire-fighting substance mixed with seawater. It may also be used in training, testing and maintenance. Source: 40 CFR 1700.4.



Foreign-flagged ships

Our fleet

- Use of AFFF in US territorial waters is limited to emergencies only
- Every use must be recorded
- Discharge for training, testing or maintenance purposes is not permitted
- Plan drills involving foam for waters outside the permit area



US-flagged ships

For reference

- Training and maintenance discharges may be made
- The foam must be collected on board
- The collected foam is delivered to shore terminals
- The record requirement still applies

Firemain systems and small boat wet exhaust



2.2.12 Firemain systems

- Discharges from the fire-fighting system are made only in an emergency
- Discharges using this system in port are reduced as far as possible
- Except in an emergency, no discharge is made in the areas listed in VGP section 12.1
- Washing anchor and chain with the firemain must comply with the requirements of 2.2.8



2.2.21 Small boat engine wet exhaust

- Such engines are included in the planned maintenance system
- Maintenance is carried out in accordance with the manufacturer instructions
- Low sulphur fuels should be used in these engines wherever possible

Welldeck discharges

The welldeck is the opening at the waterline where smaller and amphibious craft can dock and be supplied. It is mostly found on military ships. Source: 40 CFR 1700.4.



Graywater in the welldeck

Puddles containing graywater must not be discharged except in an emergency.



After washing craft

The puddles that appear after washing vehicles and equipment must never contain litter or oil.



Housekeeping

Keep the space and its surroundings clean and tidy so that no rubbish or debris can be washed out.

Anti-fouling hull coatings



Certificate on board

The certificate for the anti-fouling paint used on the ship must be held on board.



FIFRA or US standard

The anti-fouling paint must meet FIFRA or the applicable US standard.



No TBT

There must be no tributyltin (TBT) in the paint content.



Sealed TBT coatings

If a TBT coating has been applied, the certificate must state that an effective sealer coat has been applied over it.



Evidence of permissibility

The ship must hold a certificate, or a written permission, showing the coating is usable under the Vessel General Permit.

Cathodic protection

- Anodes with the least toxic content available will be used.
- Excessive numbers of single-use sacrificial anodes should not be fitted.
- Cleaning or renewal is carried out while the ship is in the shipyard, to reduce excessive scaling and flaking from oxidised anodes or from the dielectric coating of the ICCP system.
- A certificate must be obtained from the shipyard covering the cleaning of the cathodic protection and the maintenance carried out.
- On large ships, galvanic anodes cannot deliver the current economically required for full protection.
- Impressed current cathodic protection (ICCP) systems use anodes connected to a DC power source, normally through a converter that turns the AC supply into DC.



Alternative power sources

An ICCP power supply does not have to come from the main switchboard. Solar panels and thermoelectric generators can both be used as the DC source, which matters on unmanned or laid-up units.

Sonar dome discharge and oil-to-sea interfaces



2.2.22 Sonar dome discharge

- The anti-foulant on the sonar dome or echo sounder dome is washed and treated
- Seawater or fresh water held inside the sonar dome must not be discharged in US territorial waters
- Hold the water and land it ashore



2.2.9 Underwater seals, wire ropes and submerged equipment

- Seals are kept in good condition and checked or renewed at the shipyard as necessary
- Shipyard maintenance reports are kept on board
- There must be no excessive oil or grease on wire ropes
- An oil boom is used against possible leaks during underwater maintenance; the response equipment must be available and the necessary resource provided by the Company

Underwater ship husbandry discharges

The discharge of wastes generated while cleaning the hull and underwater body of the ship in the water.



Major cleaning ashore

Major hull cleaning activities should be carried out in shipyards, where the waste can be contained.



In-water cleaning

If the operation must be carried out with the ship afloat, waste discharge to the environment is kept as low as possible.



Copper anti-foulants

When cleaning anti-foulants containing copper, care must be taken that no visible plume or layer forms in the water.

Additional requirements for tankers

Section 5.5 of the permit adds two items for tankers on top of the 26 general categories.



Inert gas scrubber discharge

- The scrubber must be an internal part of the IGS installation
- The IGS is maintained regularly in accordance with the planned maintenance system
- Wastes generated by the system are reduced through that maintenance



Deck washdown and runoff from rain or spray

- Scupper plugs must be fitted during cargo operations
- All oil spills on deck are cleaned with suitable cleaning materials
- The water around the ship is observed continuously for leaks
- Deck washings are not permitted in US waters
- Personnel are trained in the environmental protection procedures

Storage of materials on deck

All deck-mounted equipment exposed to rain, seawater spray and wind is treated as a potential discharge under this permit — including cargo-related items and deck equipment.



Anything that can go overboard is a waste

All objects that can fall off the deck are considered a waste under the VGP.



An effective environmental management system

The permit requires every ship operator to establish an effective environmental management system.



Minimise pollutants released

Environmental protection is achieved by minimising the pollutants released during cargo operations or during the use of ship equipment.



Shelter what is stowed on deck

The necessary precautions are taken so that materials or cargo kept on board are not exposed to rain, water spray and wind.

Substances that have no permit at all

The disposal or discharge of toxic and hazardous substances into the environment is not a matter for which permission exists.



No discharge

There is no permitted route to sea for these substances under any circumstances.



Not exposed

Such materials must not be exposed to spray, rain or wind.



Sealed and marked

They are stored safely on board in sealed and clearly marked containers.



Recorded

Any discharge procedure carried out for the safety of the ship or personnel must be recorded.

Wastes and discharges not covered by the VGP

These items are generally covered by other regulations in force, or their release to the environment is prohibited outright.

- Sewage from vessels
- Used oil
- Garbage
- Waste generated during photo processing
- Waste generated during dry cleaning
- Medical waste
- Chemical liquid wastes harmful to the environment and to life
- Tetrachloroethylene (perchloroethylene) degreaser
- Waste discharges made under other rules and permits



Out of the VGP is not out of the law

- Every item on this list is regulated somewhere else
- Sewage, oil and garbage each have their own regime
- The wrong logbook is still a non-conformity
- Ask the DPA if the correct route is not obvious

Absolute prohibitions in US territorial waters

Learn these nine. Except in a genuine emergency, none of them may be discharged inside the permit area.



Bilgewater

2.2.2 — except in an emergency.



Deck washing

2.2.1 — washing decks is forbidden.



Boiler blowdown

2.2.6 — except in an emergency.



Scrubber sludge

2.2.26 — sludge must not be discharged.



Gas turbine wash water

2.2.14 — never overboard.



Compensating discharge

2.2.16 — never in VGP waters.



Refrigeration condensate

2.2.18 — prohibited.



Sonar dome water

2.2.22 — must not be discharged.



Graywater with sewage

2.2.25 — except in an emergency.



SECTION 03

Inspections and Corrective Action

The permit is verified by self-inspection. What is found must be corrected within a stated time and written down.

03

What the watch is looking for

Watchkeeping personnel should observe the water surface around the ship at all times.



Sheen and glow on the water

A sheen on the surface is the first and most common indicator of an unpermitted discharge.



Litter, chemicals and foaming

Look for floating litter, chemical traces, abnormal colour changes, foaming and other contaminants.



Deck puddles, ballast and bilge

Special attention is paid to puddles on deck, to ballast water and to bilge water.



It is an SMS requirement too

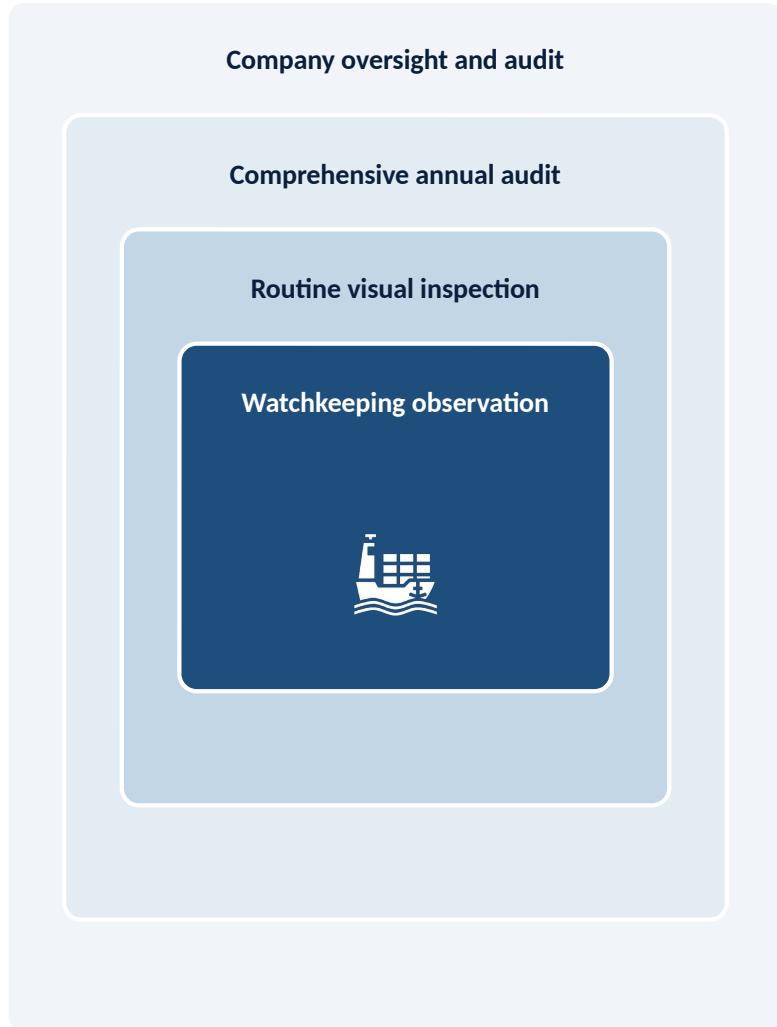
These requirements must also be met within the safety management system (SMS), not only under the permit.



Report it, do not judge it

It is the responsibility of all personnel to report water pollution to the Master or the responsible officer. Deciding whether it is reportable is not the job of the person who found it.

How VGP compliance is actually checked



Continuous. The watch observes the water around the ship and reports anything abnormal to the Master.



Routine. Decks, engine room, cargo spaces and lockers are checked periodically — at least weekly, or once each voyage.



Annual. A comprehensive audit by trained personnel of every VGP item that does not require a shipyard.



Shipyard. Hull, ballast tanks, seals and anti-fouling are examined when the ship is out of the water.



Ashore. The Company reviews the records, monitors trends and reports non-conformity to the EPA regional office.

Inspection regime at a glance

Inspection	Frequency	Carried out by	Scope
Routine visual inspection	Periodically — at least weekly, or once each voyage	The Master, or a responsible officer appointed by him	Decks, engine room, cargo spaces and lockers; garbage, exposed materials, oils and other contaminants
Comprehensive annual audit	Annually	Trained personnel — Master, chief engineer, environmental engineer, inspectors, or a person appointed by the classification society	All VGP items that do not require a shipyard, with special attention to areas that could cause pollution and a VGP violation
Shipyard inspection	At each drydocking	Shipyard and ship staff together	Hull fouling and flaking anti-fouling, TBT or organotin areas, ballast tanks, bilges, pumps, OWS sensors and protective seals

The results of the annual inspections are recorded in the management system and in the log.

The four things to look for

Decks, engine room, cargo spaces and lockers are checked periodically for the following.



Garbage

Loose rubbish and debris anywhere it could be washed or blown overboard.



Exposed materials

Materials left exposed to seawater, rain and other external factors.



Oils

Oil traces, drips and residues on deck and around machinery.



Other contaminants

Chemicals, paint, cleaning agents and anything else that could reach the sea.

The comprehensive annual audit



Trained personnel only

The audit must be carried out by personnel who have received the necessary training.



Who may carry it out

The Master, the chief engineer, an environmental engineer, an inspector, or a person appointed by the classification society.



Everything short of a drydock

The audit includes all of the items mentioned in the VGP that do not require a shipyard.



Focus on the pollution risk

Special attention is paid to the areas that could cause pollution and result in a VGP violation if discharged to sea.



Results are recorded twice

All results of the annual inspection are recorded in the management system and in the log book.

Shipyard inspection points

Attention should be paid to the following during shipyard inspections — the list is not exhaustive.

- Living organisms and flaking anti-fouling paint on the underwater hull
- Areas where paints containing TBT or organotin have been used
- Ballast tanks, including sediment accumulation
- Bilges, pumps and oily water separator sensors
- Protective seals that prevent lubricating oil and hydraulic oil leaks
- Oil and chemical preventive measures, and how effective they actually are



Findings become actions

Corrective action must be initiated wherever an item is found that could cause a violation of these permits. The result of every annual inspection is recorded in the management system and in the log.

Four kinds of deficiency requiring corrective action

Measures must be taken immediately to eliminate or prevent the recurrence of any deficiency or problem found during an inspection.



A VGP violation

Violation of one or more of the items covered by the VGP.



A change of measure needed

A need to change the administrative or management measures, identified after an audit.



Measures that do not fit

The administrative measures do not meet current quality standards, or are too tight to be met in practice.



Measures being ignored

Environmental pollution measures or effective management practices are being ignored, or the discharge of pollutants is not being reduced.

How a corrective action is handled

1

Detection

A violation is found by routine visual inspection, comprehensive annual inspection, operational inspection and evaluation by the crew, a flag State or other official, or by any other means.

2

Reporting

Violations may be reported by any member of the ship staff. Nobody needs authority to raise one.

3

Evaluation

The Master or the responsible officer evaluates the corrective action, considering the nature of the event, its causes, and the options for eliminating it.

4

Record and certify

The evaluation is then recorded, signed by the Master or his representative, and certified.

5

Close out on time

The action is completed within the deadline set for its category, and the closure is recorded.

How long you have to put it right

The time allowed depends on the source and nature of the problem. Meeting some requirements may need additional time, but problems are resolved as quickly as possible.

Category	What it covers	Deadline
Simple arrangements	Corrective action that can be completed with the staff and resources already available — cleaning and good order, reporting, record keeping, control of displays, and certain operating and maintenance activities	As soon as possible and not exceeding 2 weeks. The problem must be eliminated before the ship re-enters US territorial waters. The time limit carries no tolerance
New equipment and parts	Situations where corrective action requires new equipment or spare parts	Causes identified and resolved within 3 months of the problem arising; or before the ship next enters US waters if it leaves and returns within 3 months. If repairs cannot be made within 3 months this must be stated in writing and completed as soon as possible thereafter
Shipyard	Corrective action requiring major or extensive renewal, modification or repair, which can only be carried out in a shipyard	The root causes must be corrected at the first shipyard

WHY THE TWO-WEEK LIMIT HAS NO TOLERANCE

Every day a violation continues is assessed separately

Violations of the rules are evaluated separately for each day they persist, and continuation of a violation causes the penalties to be doubled. A deficiency left open is not one finding — it is one finding per day.

Fix it now

Before re-entry

Record the closure

Reporting non-conformity to the EPA



At least annually

Any non-conformity found must be reported to the relevant EPA regional office on a minimum annual basis.



Choosing the office

If the ship has non-conformities in more than one region, contact the office of the region where most of them occurred, or the office of the region in which the ship trades most.



Ask the DPA

The information needed for reporting non-conformities can be obtained from the DPA.



Report and record

The report to the EPA does not replace the on-board record — both are required.

What every ship must record

All ships are required to record the following information.

- Shipowner and ship information
- Voyage records
- Violations of the permit
- Findings and corrective actions arising from routine inspections
- Analytical results of the observations made
- Findings and corrective actions from the annual audit
- All information required by the EPA for the ship
- Maintenance and discharge activities
- The approval letter and authorisation number of the NOI application



Where they live, and for how long

- VGP records may be kept in a separate book or in the existing log books
- The same record does not need to be kept in more than one place
- VGP records are kept on board for 3 years
- Records must be available for inspection at any time

What a violation actually costs

VGP-related items carry serious penalties for violation, and the individual is exposed as well as the ship.

\$10,000

Maximum fine for tampering with a device or entering false information

2 yrs

Maximum imprisonment for tampering or falsification

6 mths

Maximum imprisonment for an incorrect or altered VGP document

x2

Penalties are doubled where the violation continues



Assessed daily — Violations of the rules are evaluated separately for every day they continue.



And other US laws — Other penalties may also apply for the same violation under other United States laws.

The two offences and their penalties

Both the ship and the individual are exposed. Neither offence requires pollution to have occurred.



Tampering or false information

The more serious offence

- Replacing or tampering with a device, or with an application that has to be followed
- Entering false information in a record
- A maximum fine of \$10,000
- A maximum of 2 years in prison
- In some cases both are applied simultaneously, and continuation of the violation doubles the penalties



Incorrect or altered documents

Reports, certificates and documents

- Submitting a report, certificate or document required under the VGP incorrectly
- Submitting one after alteration
- A maximum fine of \$10,000 per violation
- Or a maximum of 6 months imprisonment, or both at the same time
- Other penalties may also apply under other US laws

Special requirements of individual states

Some US states have stricter rules in addition to the VGP.



Stricter state rules

A number of states impose requirements over and above the federal permit, particularly on ballast water and discharge reporting.



The Master must check

The Master is obliged to check the applicable state rules according to the region the ship is going to.



Where to look

For detailed information, see VGP section 6. California is the most frequently cited example of a stricter regime.



SECTION 04

Bunkering Operations

The highest-consequence oil transfer the ship performs, carried out inside the permit area with the water surface under observation.

04

Bunkering under the VGP

The necessary measures will be taken against leaks and spills in all bunkering operations, and effective management practices will be followed through to the end of the operation.



Prepare against leaks and spills

Scupper plugs, drip trays, absorbents and response equipment are in place before the transfer begins.



Immediate intervention

If there is a leak or an overflow, the necessary intervention will be made immediately.



Never fill a tank completely

Fuel tanks are never filled completely — a safety margin is always left in the tank.



Watch interconnected tanks

Where fuel tanks are interconnected, the necessary measures must be taken to prevent those tanks from overflowing.



Observe the water throughout

The water around the ship is observed continuously for any sign of a leak while the transfer is in progress.

Bunkering do and do not

✓ Do

- Fit and check scupper plugs before the hose is connected
- Keep absorbent materials and a spill kit at the manifold
- Agree the transfer rate, topping-off rate and stop signal in advance
- Keep a continuous watch on the water surface around the ship
- Record the operation and any drip, spill or intervention

⊘ Do not

- Fill any fuel tank completely
- Leave interconnected tanks unattended during topping off
- Wash a spill overboard rather than collecting it
- Assume the previous watch checked the scuppers
- Delay reporting a spill while deciding how large it is

Filling lifeboat and rescue boat fuel tanks

These procedures cover the filling of the fuel tanks of lifeboats and rescue boats — small quantities, but inside the permit area and in full view.



Nothing to the environment

During the transfer it must be ensured that no fuel reaches the environment.



Know the capacities

Knowing the capacity of each fuel tank helps prevent overflows. Do not fill beyond the safety margin.



Collect spills and drips

Collect spills and drips with absorbent materials, and dispose of the used absorbent as waste.



Check frequently

Lifeboats and rescue boats are checked frequently for leaks and for damage.



Trays underneath

Keep a drip tray under the filling point and under the engine while the boat is stowed.



Record it

Record the filling operation, the quantity and any spill, however small.

Spill response essentials

A spill inside US waters is a permit violation the moment it reaches the sea. Speed decides the outcome.



Stop the transfer

Stop the operation and close the valves before anything else.



Contain it

Deploy absorbents and containment. An oil boom is used where a leak may reach the water.



Raise the alarm

Inform the Master or the responsible officer at once, and follow the Vessel Response Plan.



Record everything

Time, position, quantity, cause and the action taken all go into the record.

Training requirements for bunkering personnel

All personnel responsible for bunkering should be trained in preventing environmental pollution, preventing human error, responding to a leak or overflow, and using the related equipment effectively.

- Officers and senior staff who control and manage these operations receive comprehensive training including the items in VGP section 2.1.
- 2.1.3 — leaks and spills.
- 2.1.4 — transfer of oil and petroleum-derived liquids over the side.
- Effective management practices applicable to the operation.
- 5.5.2 — additional permit requirements, including scupper plugs and intervention in a leak.
- 5.5.3 — additional inspections, including observation of the water surface and reporting.



Two levels of training

Everyone involved in the transfer needs the practical training. The officers who control and manage the operation need the full VGP section 2.1 content in addition, because they sign for the operation.

What officers and responsible personnel must also know



Vessel Response Plan (VRP)

The plan itself, and the role each person holds within it.



SMS environmental procedures

The environmental procedures of the safety management system as they apply to fuel transfer.



Duties under the SMS

The duties and responsibilities of each person according to the safety management system.



Fuel transfer procedures

The procedures relating to fuel transfer, including checklists and the pre-transfer conference.



Watchkeeping procedures

Watchkeeping procedures, which must also be displayed as a poster in various parts of the ship.



Pollution response equipment

The locations of the equipment used to respond to environmental pollution, the instructions for its use, and the emergency procedures to follow in case of a leak.



SECTION 05

Ballast Water Management

Ballast water is category 2.2.3 of the permit, and it carries more record-keeping than any other discharge on board.

05

Which rules apply to ballast in US waters

On ships that carry ballast and trade to US territorial waters, records must be kept properly and regularly.



33 CFR

Records are kept in accordance with the procedures of 33 CFR and with the ship own ballast water management plan.



CWA section 311

Ballast operations are also covered by section 311 of the EPA Clean Water Act.



State rules

Some states have tighter ballast procedures and may require different reporting — California is the example most often encountered.



33 CFR 155

USCG requirements under 33 CFR 155 apply to the ballast record book referenced by VGP 2.2.3.

Areas to avoid when taking ballast

Ballast must not be taken in areas that may affect natural life, in protected areas or over coral reefs. Avoid ballasting, or take as little ballast as possible, in the areas and situations below.

- In areas known to contain harmful organisms and pathogens
- In places where sewage is discharged to the sea
- In areas where underwater dredging or excavation is being carried out
- Where currents cause turbidity in the water
- Where organisms living on the seabed rise towards the surface at dusk
- In shallow water, where the ship propellers will lift bottom sediment into the water column
- At the crossing point of large currents
- Anywhere the ship own operations have already disturbed the bottom



The uptake decides the discharge

Everything that later has to be exchanged, treated or held was taken in at the pump. Choosing the position and the moment of uptake is the cheapest control available.

Sediment control and minimising discharge



Sediment in the tanks

- Sediment accumulated in the tanks of ships with clean ballast tanks is removed in port or in the shipyard, under control
- Sediment produced by ballast tank cleaning cannot be discharged to the sea in US territorial waters
- Tank cleaning and sediment removal are planned for the drydocking



Volume discharged

- Ballast discharge to sea will be minimised as far as possible on ships operated in US territorial waters
- Plan the cargo operation so that the ballast discharged inside the permit area is the smallest quantity the operation allows
- Use a shore reception facility where one is available

Additional ballast records for US waters

If ballast is to be discharged in US waters, or given to a shore facility, the following records must be kept.

- The source of the ballast water — date, place stated as latitude and longitude, volumes and temperatures.
- If the ballast has been exchanged, the list of ports and locations where the ballast discharged during that exchange had been taken.
- If an exchange has been made — the time, location, volume and percentage of the ballast replaced.
- The water depth at the position where the exchange was carried out.
- The estimated time, location, volume and salinity of the ballast water to be discharged to sea or given to the facility in US waters.
- Every entry is signed and retained with the VGP records.



The plan and the crew

- All owners and operators must prepare a ship-specific ballast water management plan
- The crew must be aware of how that plan is implemented
- The plan is the reference for every entry in the record
- An unfamiliar crew is a finding in itself

Discharge ashore, or treat before discharge



Transfer to a shore facility

- Where a system has been installed to transfer ballast to a shore facility, the owner or operator should use it when necessary
- Port facilities required for ballast processing will offer this service economically
- Record the volume transferred and the facility used



Treatment before discharge

- Ballast filtering and treatment systems approved by the US Coast Guard must be used if ballast is discharged to sea
- The approval documentation is held on board
- The system is operated and maintained according to its manual, and the operation is recorded

Ballast water exchange requirements by trade

The exchange requirement depends on the route and on where the ballast will be discharged.

Trade	Applies to	Exchange requirement
Ocean voyage	Ships navigating the ocean, ballasting at the maximum distance from the coast, navigating 200 miles from the nearest land and outside the exclusive economic zone, and intending to discharge ballast in permitted waters	Ballast must be exchanged outside the exclusive economic zone and 200 miles from the nearest land, and the exchange should be done as early as possible in the voyage
Pacific coast	Ships navigating on the Pacific coast that ballast in waters 50 miles from the coast and will discharge in territorial waters subject to this permit	Exchange at a minimum distance of 50 miles from the coast, in a minimum water depth of 200 metres. If the ship will navigate 200 miles from the coast during the voyage, the exchange is carried out then
Atlantic coast	Ships navigating the Atlantic coast that will not leave the US exclusive economic zone	No ballast water exchange is required

The distances that decide a ballast exchange

The exchange position is fixed by these figures. If the ship cannot reach them, the exchange cannot be counted.

200

Miles from the nearest land, and outside the exclusive economic zone

50

Miles minimum from the coast for a Pacific coast exchange

200 m

Minimum water depth for a Pacific coast exchange



As early as possible — The exchange is carried out as early in the voyage as conditions and safety allow.



Atlantic coast — Ships on the Atlantic coast that will not leave the US exclusive economic zone are not required to exchange ballast.



Record the position — Time, location, volume, percentage exchanged and water depth all go into the ballast water record book.

Vessels arriving with empty or residual tanks

Ships arriving from international waters that have reported no ballast on board, or that have one or more empty ballast tanks or unpumpable residues in the tanks, must apply one of the following.



Seal the tanks and valves

Seal the ballast tanks and their associated valves so that no ballasting or deballasting can take place in US waters.



Or flush the tanks

Alternatively the ballast tanks are flushed with seawater at the minimum required distance outside US territorial waters.



Record whichever route is used

The sealing or the flushing is recorded in the ballast water record book with time and position.



Expect it to be checked

A declaration of no ballast on board does not remove the tanks from the scope of the inspection — residues carry organisms too.

Deciding and recording a ballast exchange

1

Start from the plan

Open the approved ballast water management plan and confirm what the voyage requires.

2

Identify the trade

Ocean voyage, Pacific coast or Atlantic coast — each carries a different exchange requirement.

3

Choose the position

Outside the exclusive economic zone and 200 miles from the nearest land; or a minimum of 50 miles off the Pacific coast in at least 200 metres of water.

4

Exchange as early as possible

Carry out the exchange as early in the voyage as conditions and safety allow, rather than leaving it until the approach.

5

Record and report

Enter source, position, volume, percentage exchanged, depth and salinity, and report as the state and the permit require.

When the exchange may be omitted

Where the Master decides that the ballast operation is not safe, he may omit it — provided the decision is recorded.



Safety of the ship

The Master may decide the exchange is unsafe and not carry it out, by entering it in the log book and record book with the date, place and reason.



Approved alternative system

The exemption also applies where the Master uses a system approved by the USCG as environmentally sound in place of the normal arrangement — 33 CFR Ch. 15.



Ballast never discharged

No exchange is required where the ballast is retained in the tanks and is not discharged to sea at any time.



No international voyage

No exchange is required where the ship does not undertake international voyages and does not leave the waters subject to the permit.



Always write it down

In every case the Master records the transaction, stating the date, the place and the reason.



Not a routine option

The exemption is a safety decision, not a scheduling convenience. Repeated use will be questioned.

Ballast water management training

All owners and operators must ensure that the Master, the operator and the personnel responsible for the ballast tanks receive all training on ballast water management. The obligation can be met by training in the effective management practices (BMPs) for ballast operations.



The plan and its records

The approved ballast water management plan and the keeping of the relevant records.



IMO and US regulations

International (IMO) and United States ballast water regulations, including the state requirements that apply.



SMS environmental procedures

The environmental procedures of the safety management system, and the duties and responsibilities assigned under it.



Watchkeeping procedures

Watchkeeping procedures, and awareness of the pollution that a ballast operation can cause.



Response equipment

The locations of the equipment used for response to pollution, and the instructions for its use.



Emergency procedures

The emergency procedures to follow against pollution arising from a ballast operation.



SECTION 06

Notifications and Office Duties

Coverage under the permit is obtained, transferred and ended by notification to the EPA — all of it electronically.

06

Obtaining coverage under the permit

A Notice of Intent (NOI) must be sent to the Environmental Protection Agency to obtain the permits required for US territorial waters.



Notice of Intent categories and timing

Category	Time of NOI	Date of entry into force for the ship
Transfer of a ship to another business or shipowner	At the time of transfer	The transfer date, or the time at which the new EPA authorisation is received
Ships not previously covered by the VGP	30 days	30 days after the Notice of Intent is submitted to the EPA
New ship purchase	30 days	30 days after the Notice of Intent is submitted to the EPA

The ship cannot enter US territorial waters until coverage has taken effect.

The EPA eNOI system



Everything is electronic

The NOI, the Notice of Termination and the annual report are all submitted through the EPA eNOI system.



Vessel search is public

The eNOI system provides a public search for any vessel that has submitted an NOI, a Notice of Termination or an annual report under the VGP.



Results are downloadable

Search results are presented as a table and can be downloaded as a spreadsheet or an HTML file for further review.



Assume it will be checked

Because the record is public, a port State or a charterer can confirm the VGP status of the ship before she arrives.



The address

<http://www.epa.gov/npdes/vessels/eNOI> — used for the NOI, the NOT and the one-off report. The office submits on behalf of the ship, but the data from the ship has to be right first.

Leaving the scope of the permit

All ships that will be removed from the scope of the VGP declare it with a Notice of Termination (NOT) to the EPA.



When to submit a NOT

This procedure is followed when the intention is to leave the scope of the permit completely — for example on sale out of US trading or on withdrawal from service.



When not to

The declaration does not need to be made every time the ship sails out into international waters. Ordinary trading does not end coverage.



Transfer instead

Active permits can be transferred to another shipowner or operator, which is normally the correct route on a sale.



How to submit

The NOT is submitted electronically at <http://www.epa.gov/npdes/vessels/eNOI>

The one-off report to the EPA



Between month 30 and month 36

Owners or operators must submit a one-time report between the 30th and the 36th month after the start of the permit, for each ship subject to the permit.



Ship first, office second

After the personnel on board have completed the parts relating to the ship, the office sends the report electronically to the EPA.



Why the EPA wants it

The purpose of the report is to allow the EPA to follow how VGP practices are being applied on board.



What it draws on

The report is compiled from the on-board VGP records — inspections, findings, corrective actions and discharge activity.



SECTION 07

Duties of the Ship Personnel

The permit is kept by the people who wash the deck, stow the drums, paint the ship side and wash the dishes.

The standard of watchkeeping under the VGP



Observe the water at all times

The watchkeeper observes the water surface around the ship continuously for sheen, litter, chemicals, abnormal discolouration, foaming and other contaminants.



Where the problem usually starts

Special attention is paid to puddles on deck, and to ballast and bilge water on board.



Everyone reports

It is the responsibility of all staff to report water pollution to the Master or the responsible officer.



What is seen is recorded

Observations, and any pollution found, are entered in the record so that the annual report can be compiled from them.



The first minute matters most

A sheen alongside can still be contained. The same sheen an hour later cannot. Report immediately and let the Master decide what it is.

Cleaning and stowage in US territorial waters

While the ship is in the US territorial sea, all personnel are obliged to ensure good order in the following matters.

- No material is allowed to fall or be swept into the sea, or to be left exposed to rain and spray.
- All materials and equipment are kept in closed lockers.
- Substances that could enter the sea are not allowed to come into contact with oil or petroleum-derived substances.
- Hazardous and toxic substances are kept in the dedicated, sheltered lockers built into the ship for that purpose.
- Deck equipment left in the open is covered or moved into a locker before the ship arrives.
- Anything that could go overboard is treated as a waste under the permit, whatever it is.



It applies to cargo too

Deck-mounted equipment and cargo-related items exposed to rain, seawater spray and wind are potentially a waste within the scope of these permits. The rule is not limited to the stores of the ship.

Five requirements for every container on deck



Suitable and closed

Made of suitable material and tightly closed.



Labelled

Clearly labelled with the contents.



Secured

Securely lashed and unable to shift.



Safety margin

Not filled beyond the safety margin.



Sheltered

Stowed with minimal exposure to rain and spray.

Accumulation on deck

While the ship is in port, the relevant personnel will reduce as far as possible the accumulation on deck that could reach the sea.



Permitted only when

- Effective management practices are being followed
- The accumulation contains no contaminant — no garbage, no oil
- Drip pans are kept under deck equipment where necessary
- Pans and trays are emptied to the waste containers and cleaned frequently
- The discharge is recorded if it causes any pollution at all



Not permitted

- Allowing accumulation to run off before checking what is in it
- Hosing deck accumulation over the side in port
- Leaving drip pans full until they overflow
- Sweeping debris into the scuppers
- Treating rainwater as automatically clean

Material that must be kept out of the sea

On deck and on the parts of the ship above the water surface, the discharge of the following to the sea must be reduced.



Rust

Rust scale from decks, rails, fittings and the ship side.



Cleaning agents

Detergents and chemical cleaners used on deck.



Paint chips

Paint crumbs and flakes from chipping and preparation.



Non-slip material

Fragments of non-slip deck coverings and their backing.

Painting inside the permit area

When painting, the following points must be observed so that paint does not reach the sea.



Do

- Choose calm conditions and shelter the work area
- Collect chippings and preparation debris before painting begins
- Keep containers closed and stowed away from the ship side
- Take care not to over-apply so that paint runs off the surface
- Plan ship side work for outside US territorial waters



Do not

- Spray paint in windy weather
- Paint the ship side or the boards in US territorial waters
- Leave open paint tins unattended on deck
- Wash brushes and rollers over the side
- Allow overspray to drift across the ship side

Using the fire-fighting system in port



Minimise the discharge

While the ship is in port, discharges from the fire-fighting system are reduced as far as possible.



Flushing the deck

The fire line may be used to flush the deck with a portable pump or the fire pump where the supply is seawater — provided no pollution is caused and the VGP rules are not violated.



Washing the anchor chain

Washing the anchor chain with this system is not against the rules, provided it complies with the VGP requirements of 2.2.8.



Systems kept pressurised

Some ports require the main fire line to be kept pressurised and running. In those cases the water reaching the sea is reduced as far as possible.



Hydrant-activated systems

The same consideration applies to systems activated by opening a hydrant — discharges to the sea are minimised.



Record any pollution

If a discharge to sea has caused pollution, it must definitely be recorded.

Holding capacity decides the rule



If the ship can hold graywater

- Graywater generated on board is reduced as far as possible while in port
- If the ship is able to accumulate graywater in its own tanks, it cannot be discharged in the waters subject to the permit
- The graywater is landed ashore or discharged outside the permit area



If the ship has no holding capacity

- Graywater may be discharged when the ship is under way
- The ship must be at least 1 mile from the coast
- Only phosphate-free, non-toxic soaps and detergents are used
- The discharge is recorded

What to watch in the galley

The galley is the single largest contributor to graywater quality, and the easiest place to improve it.



Keep cooking oils out

The quantity of oils used in meals reaching the graywater system is reduced as far as possible. Cooking fats are never drained into graywater.



Clean the dishes first

Dishes are cleaned of food residue as far as possible before rinsing, to reduce the food waste reaching the graywater system.



Check the product

Soaps and detergents used in the galley must contain no phosphate, no toxic compounds and nothing that can harm living organisms.



How to recognise them

Such products are generally branded as environmentally friendly, organic, or harmless to the environment.



Buy the right thing

This has to be considered at the requisition stage — the wrong detergent cannot be corrected once it is on board.



Separate the waste

Food and oil residues go to the waste stream, not to the sink.

THE WHOLE PERMIT IN ONE LINE

If it can reach the water, it is your discharge

Under the Vessel General Permit every object, residue and puddle on deck is a potential discharge, and every discharge belongs to somebody. Minimise it, contain it, record it — and report it if it goes wrong.

Minimise

Contain

Record

Report



Clean water, clear records

The VGP asks for no new equipment — only for the ship to be operated and maintained so that what leaves her is as clean and as small as it can be made, and for the proof to be written down.



Watch the water around the ship



Know your prohibited discharges



Close findings within 2 weeks



Keep records for 3 years